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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Fifth Semester B. Tech (CBCS) Degree Examinations
December 2025

BT2107-1 – UPSTREAM PROCESSING TECHNOLOGY
(For BTY)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five** full questions choosing **One** full question from **Unit – I** and **Two** full questions from **Unit – II & Unit – III** each

PART - A: MULTIPLE CHOICE QUESTIONS**20 Marks**

1. What is the primary aim of isolation of cells?
A) To reduce cell viability
B) To separate dead cells from living cells
C) To produce a suitable/useful product
D) To destroy unwanted microorganisms
2. Which enzyme is commonly used for enzymatic isolation of plant cells?
A) Trypsin
B) Pectinase (Macerozyme)
C) Collagenase
D) Hyaluronidase
3. In plant tissue culture, a higher auxin to cytokinin ratio results in:
A) Callus formation
B) Embryo abortion
C) Shoot development
D) Root development
4. Which of the following best describes a “callus”?
A) A cluster of differentiated cells with specific function
B) A layer of dead animal tissue
C) A suspension of single microbial cells
D) A mass of unorganized, undifferentiated plant cells
5. Which is the most commonly used method for sterilizing microbiological media?
A) Dry heat sterilization
B) Radiation sterilization
C) Steam sterilization
D) Gas sterilization
6. At what temperature and pressure is standard autoclaving done?
A) 121°C, 15 psi
B) 110°C, 10 psi
C) 150°C, 25 psi
D) 100°C, 1 atm
7. Which property is NOT a criterion for a good fermentation medium?
A) Maximum production of undesirable products
B) Maximum yield of product
C) Minimal problems during sterilization
D) Consistent quality year-round
8. Which has higher energy yield per gram compared to carbohydrates?
A) Vitamins
B) Oils and fats
C) Myo-inositol
D) Amino acids
9. Which medium type has all ingredients in known quantities?
A) Undefined medium
B) Defined medium
C) Complex medium
D) Nutrient medium
10. Which selective medium allows only Gram-negative bacteria to grow?
A) Blood agar
B) Potato dextrose agar
C) Eosin Methylene Blue agar
D) Nutrient agar

11. Which approach studies the effect of one factor at a time while keeping others constant?
 A) Plackett-Burman design B) Response surface methodology
 C) OFAT (One Factor At a Time) D) Factorial design
12. Which statistical tool is used to check the significance of factors in medium optimization?
 A) ANOVA (Analysis of Variance) B) Chi-square test
 C) Correlation D) Regression coefficient only
13. Which is *not* a criterion for a good inoculum?
 A) Retains product forming capabilities B) Free of contamination
 C) Healthy and active state D) Longer lag phase in fermentation
14. In bacterial processes, inoculum is usually transferred during:
 A) Death phase B) Logarithmic phase
 C) Lag phase D) Stationary phase
15. Standard steam sterilization of pipelines for inoculation is usually done at:
 A) 110°C, 60 min B) 121°C, 20–30 min
 C) 100°C, 15 min D) 134°C, 5 min
16. Which is an important aspect of aseptic inoculation?
 A) Increasing lag phase B) Cooling faster
 C) Avoiding froth formation D) No contamination during inoculation
17. The minimum inoculum density needed for growth in suspension culture is called:
 A) Critical initial cell density B) Maximum viable density
 C) Logarithmic threshold D) Stationary concentration
18. Which method is used for synchronizing plant cells in suspension?
 A) Addition of sucrose B) Cold treatment
 C) Continuous shaking D) High aeration
19. Primary cell culture is obtained from:
 A) Fresh animal tissue B) Continuous cell lines
 C) Tumor cultures D) Suspension cells
20. Anchorage-dependent cells grow best on _____
 A) Blood plasma B) Liquid
 C) Solid, inert surfaces D) Hormone-coated glass

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) How can plant cells be obtained from intact plant parts using mechanical and enzymatic methods?	08	L*2	1	2,3
	b) Differentiate between finite and infinite cell lines.	08	L2	1	2,3
2.	a) What are the two types of calluses? Write a note on different pigmentations in callus	08	L2	1	2,3
	b) How would you separate non-viable animal cells from viable ones? Describe the process with a neat labelled diagram	08	L2	1	2,3
Unit – II					
3.	a) What are the key criteria to consider when formulating an effective fermentation medium?	08	L2	2	2,3
	b) What is the importance of sterilization in industrial fermentation processes?	08	L2	3	2,3
4.	a) Elucidate the function of nitrogen sources in fermentation media, highlighting the differences between organic and inorganic nitrogen sources	08	L2	2	2,3
	b) How Del factors are calculated for heating, holding, and cooling periods during design of batch sterilization process?	08	L1	3	2,4

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| 5. a) | What are the differences between defined and complex media? Give examples of each. | 08 | L2 | 2 | 2,3 |
| b) | Analyze the process of filter sterilization of liquid media, focusing on the mechanisms involved in particle removal. | 08 | L2 | 3 | 2,4 |

Unit – III

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|-------|--|----|----|---|---|
| 6. a) | What are the criteria for inoculum development? Why must inoculum be in an active, healthy state before transfer to production fermenters? | 08 | L2 | 4 | 2 |
| b) | Why is <i>Agrobacterium rhizogenes</i> infection used for hairy root cultures, and what genetic elements are involved? | 08 | L2 | 5 | 2 |
| 7. a) | How is inoculum developed for yeast-based brewing fermentations? | 08 | L2 | 4 | 2 |
| b) | Explain growth phases of animal cells in suspension culture. | 08 | L2 | 5 | 2 |
| 8. a) | How is inoculation performed from one plant fermenter to another by maintaining pressure difference? | 08 | L2 | 4 | 2 |
| b) | Why is pH control critical in animal cell cultures, and how is it achieved? Give details. | 08 | L4 | 5 | 2 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome.
